

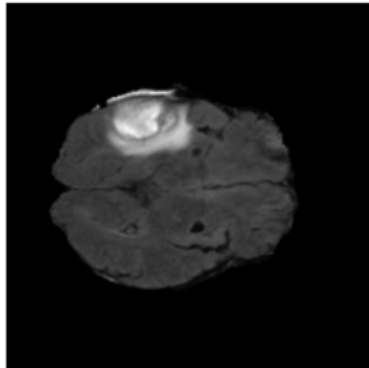
BRAIN TUMOR SEGMENTATION WITH 2D U-NET

Semantic Segmentation on BRATS Dataset

DATASET OVERVIEW

- Dataset: BRATS (Brain Tumor Segmentation Challenge)
- Each scan is a 4D volume: (H, W, D, 4 modalities)
- Modalities: FLAIR, T1, T1 Gd, T2
- Ground truth includes multiple tumor regions: edema, enhancing core, necrosis
- Big dataset ~ 8GB (zipped); unzipped ~70GB

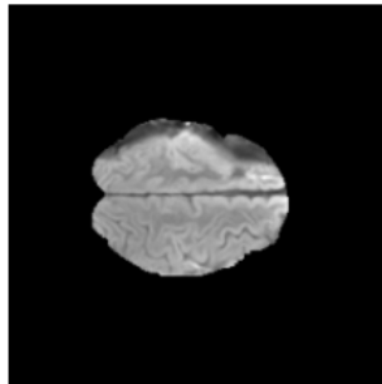
Image



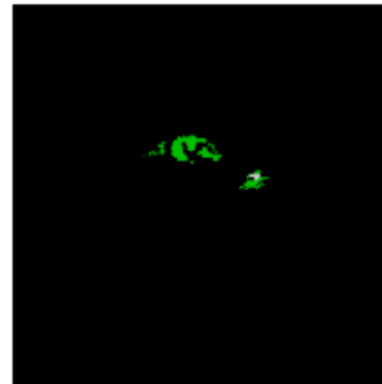
Mask



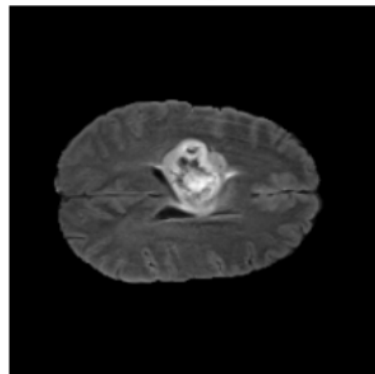
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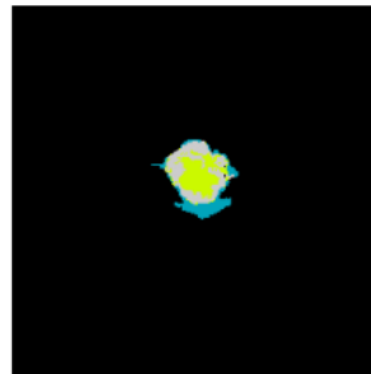
Mask



Image



Mask



TASK DEFINITION

- Problem Type: Semantic segmentation
- - Input: 2D axial MRI slices
- - Output: Pixel-wise label masks (4 classes)
- - Objective: Accurately localize and classify tumor subregions

PREPROCESSING STEPS

- Loaded .nii.gz files and extracted 2D slices
- Used only tumor-containing slices for training
- Normalized each slice (mean 0, std 1)
- Resized to 256x256 (required for U-Net with pretrained encoders)
- Saved as .npy for fast disk access
- Split dataset by volume to avoid data leakage
- Applied augmentation for training

DATA AUGMENTATION (TRAINING ONLY)

- Horizontal and vertical flips
- Random rotations (90 degrees)
- Brightness and contrast adjustment
- Gaussian blur for generalization

WHY 2D U-NET?

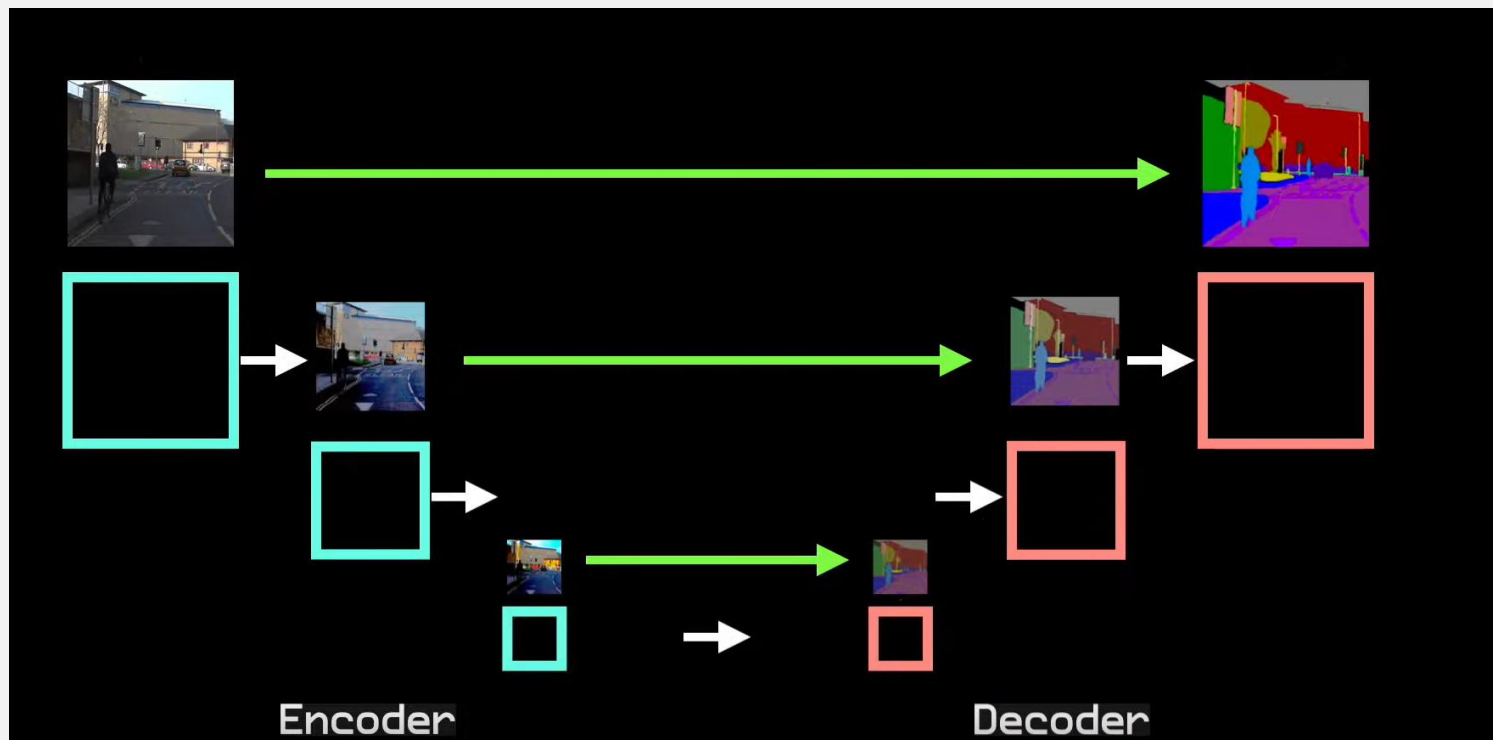
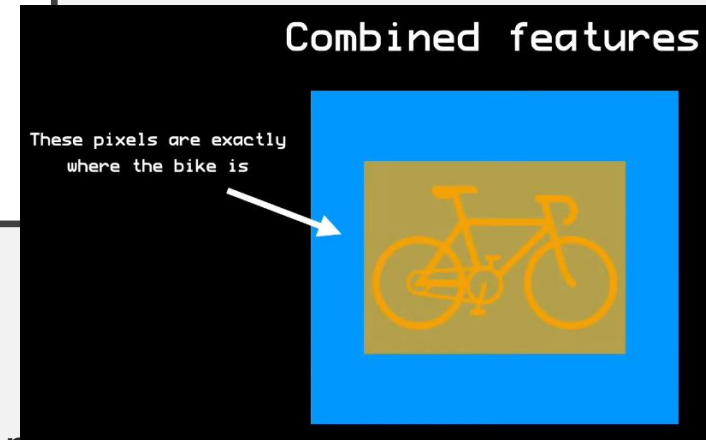
- Lower memory requirements than 3D U-Net
- Enables training on 2D tumor-containing slices
- Pretrained encoders (e.g., ResNet34) available via `segmentation_models.pytorch`
- Much faster training, good for prototyping
- Slices capture tumor features effectively in axial view

2D U-NET ARCHITECTURE

- Contracting path: convolution + max pooling to extract context
- Expanding path: upsampling + concatenation for localization
- Skip connections: combine low-level features from encoder with decoder
- Each layer

outputs feature maps with higher spatial resolution

- Final output: per-pixel class prediction (segmentation map)



TRAINING DETAILS

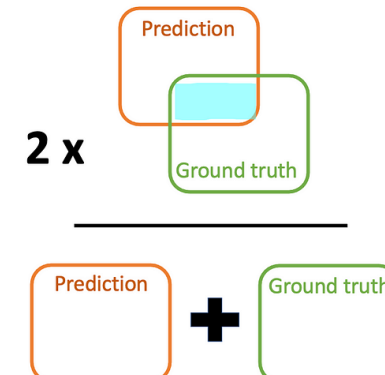
- Loss: CrossEntropyLoss
- Optimizer: Adam
- Early stopping based on validation loss
- Dice Score used as evaluation metric
- Batch size: 8, learning rate: $1e-3$
- Best model saved automatically

EARLY STOPPING

- Monitors validation loss score
- Stops training if no improvement for 5 epochs
- Prevents overfitting and wasted compute
- Custom early stopping callback implemented

VALIDATION STRATEGY

- Split dataset 70/15/15 into training, validation and test
- Test set used only for evaluation
- Dice score monitored to check generalization

$$\text{Dice} = \frac{2 \times \text{Area of overlap}}{\text{Total area}} = \frac{2 \times \text{Prediction} \cap \text{Ground truth}}{\text{Prediction} \cup \text{Ground truth}}$$


RESULTS

- Train Dice $\sim 0.85+$, Val Dice ~ 0.65
- Val Dice stagnated early $\rightarrow$ early stopping triggered
- Train/val loss gap widened $\rightarrow$ overfitting observed
- FLAIR-only input may limit generalization
- Pretrained encoder converges fast

MODEL EVALUATION METRICS

- Dice Score (per class and average)
- Pixel accuracy: $\text{correct pixels} / \text{total pixels}$
- Optional: confusion matrix to analyze misclassifications

2D U-NET (NO AUGMENTATION)

Evaluation Results:

Dice Class 0: 0.9973

Dice Class 1: 0.6467

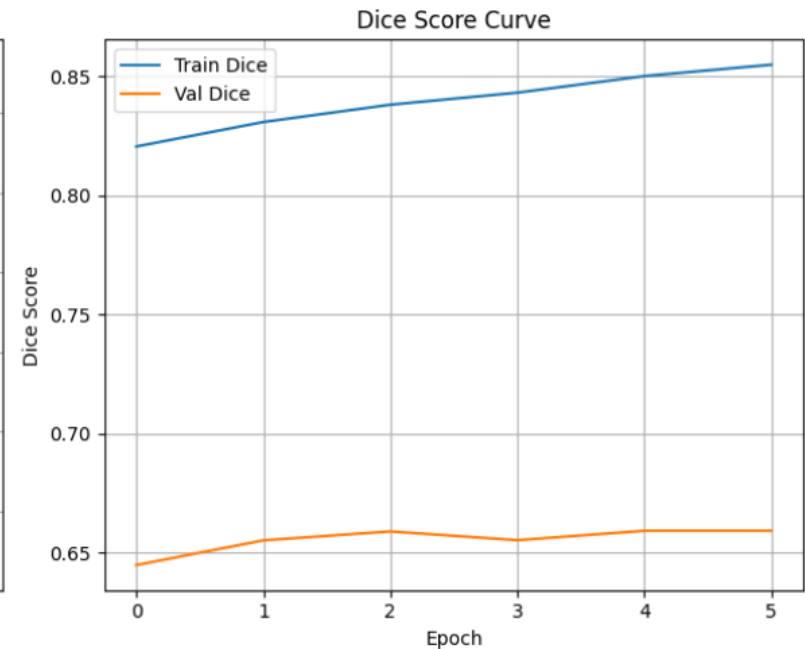
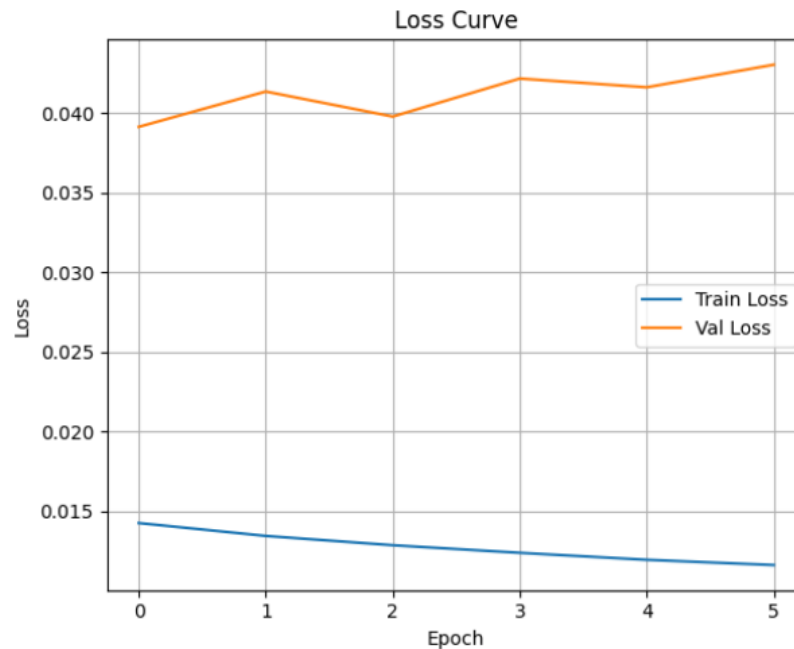
Dice Class 2: 0.4571

Dice Class 3: 0.5941

Pixel Accuracy: 0.9900

Val Loss: 0.0431, Dice: 0.6593

Early stopping triggered.



2D U-NET PRETRAINED(NO AUGMENTATION)

Evaluation Results:

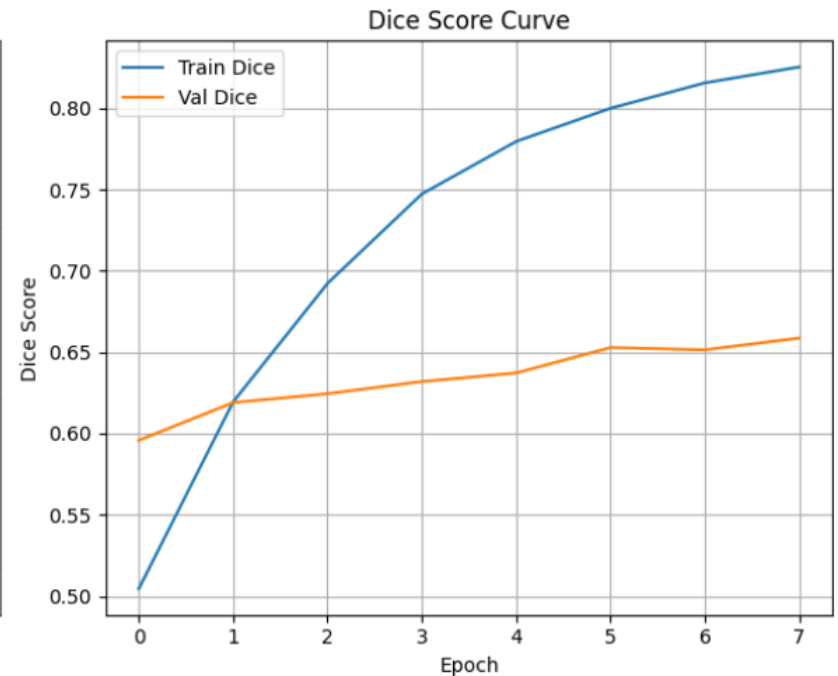
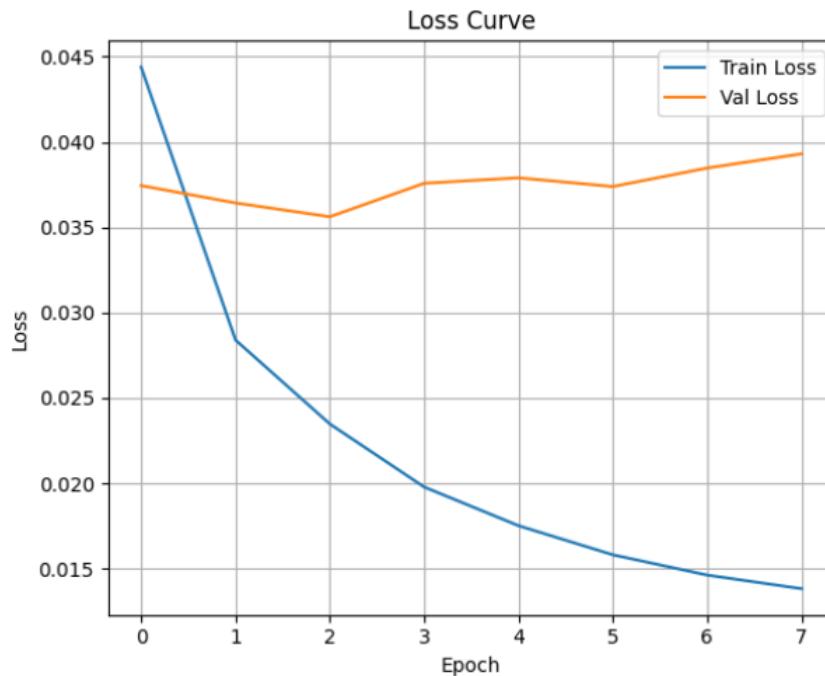
Dice Class 0: 0.9972

Dice Class 1: 0.6433

Dice Class 2: 0.4591

Dice Class 3: 0.6020

Pixel Accuracy: 0.9898



2D U-NET

Evaluation Results:

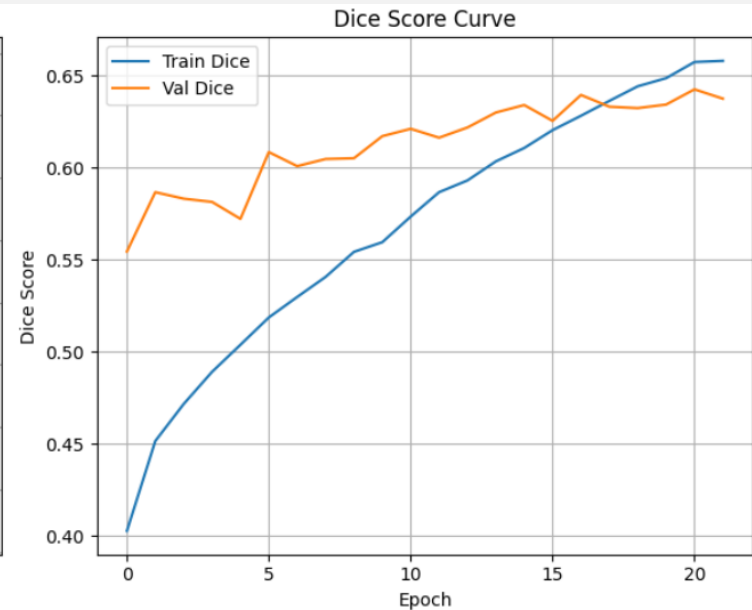
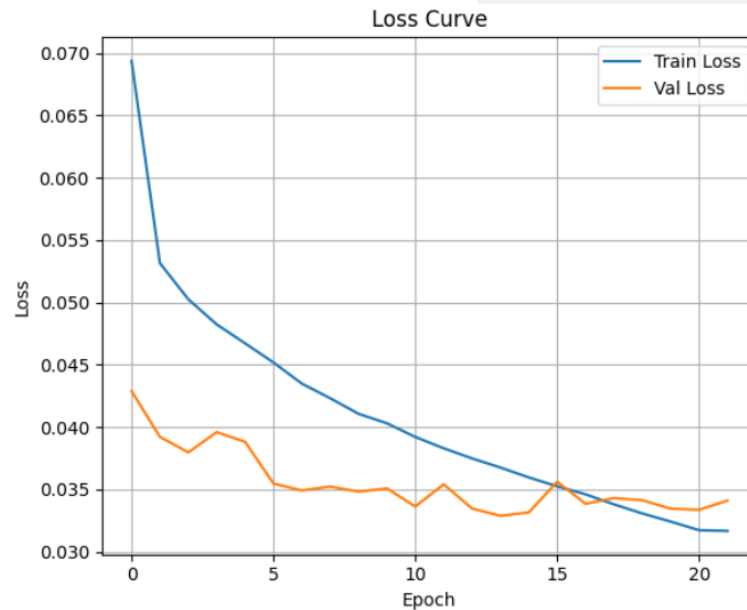
Dice Class 0: 0.9971

Dice Class 1: 0.6137

Dice Class 2: 0.4435

Dice Class 3: 0.5549

Pixel Accuracy: 0.9891



2D U-NET PRETRAINED

Evaluation Results:

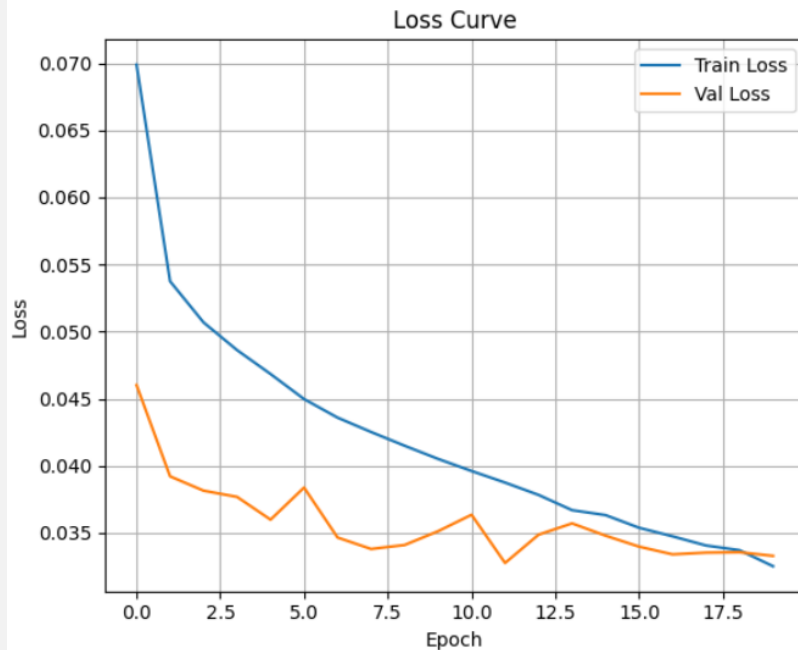
Dice Class 0: 0.9970

Dice Class 1: 0.6120

Dice Class 2: 0.4173

Dice Class 3: 0.5649

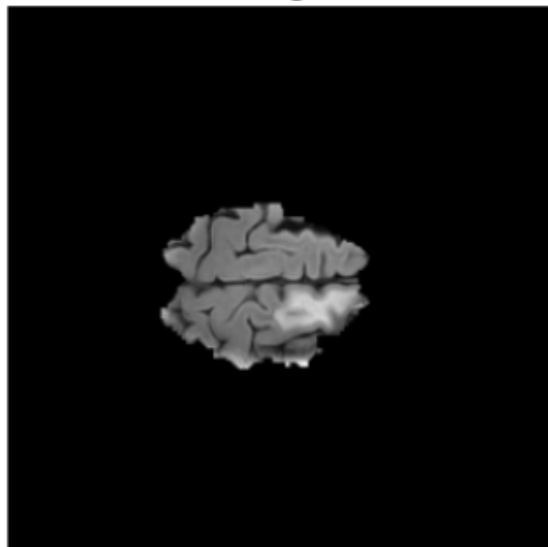
Pixel Accuracy: 0.9890



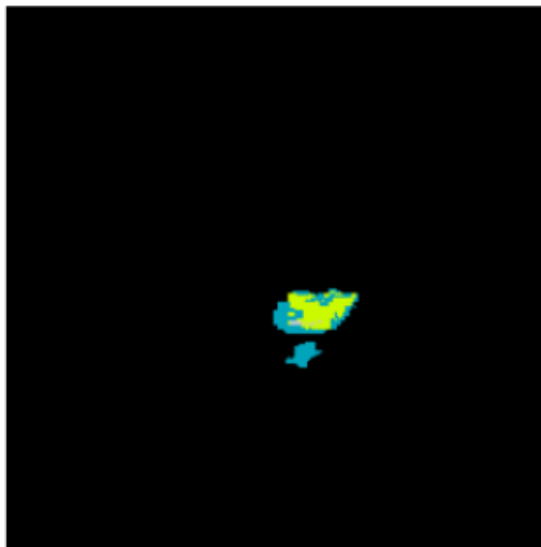
RESULTS

- The 2D U-Net with pretrained ResNet34 encoder trained well on tumor slices. However, validation loss and Dice score plateaued early, indicating limited generalization. This could be due to using only FLAIR modality, lack of tumor class diversity, or need for stronger regularization.
- Overfitting was controlled with early stopping.
- Data augmentation improved the model's learning as it could be seen in the diagrams from a few slides ago.

Image



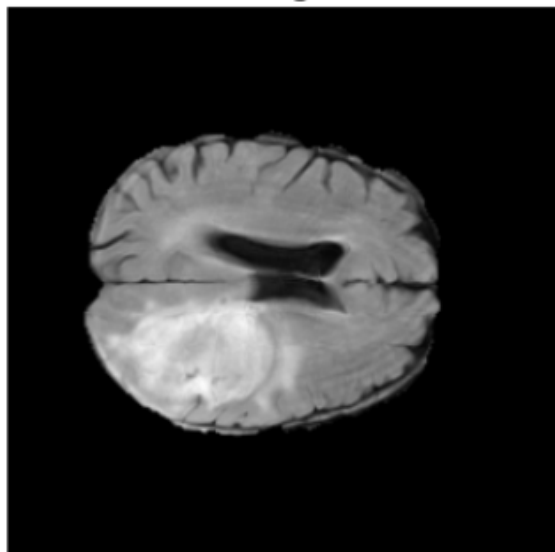
Ground Truth



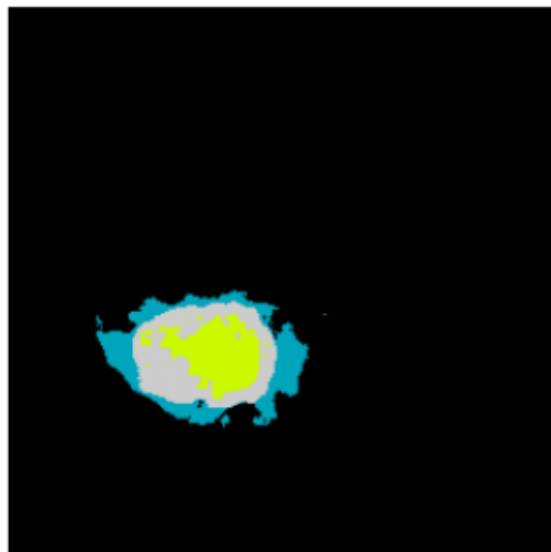
Prediction



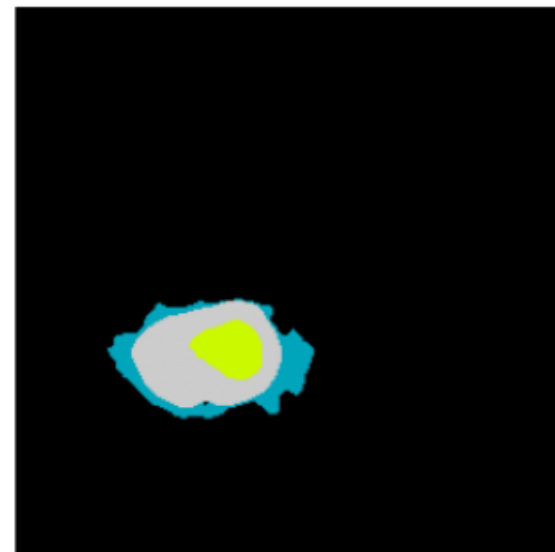
Image



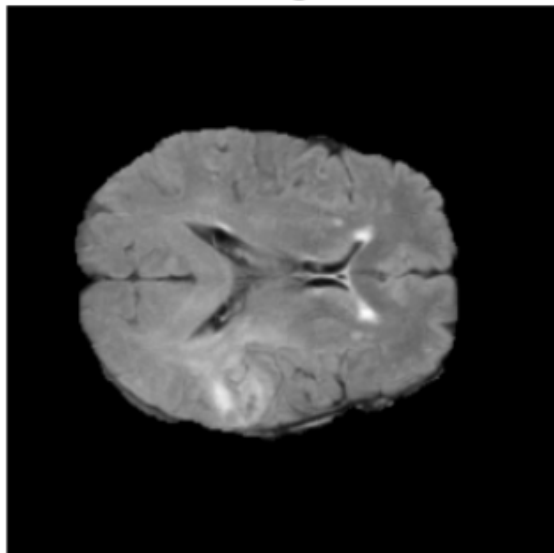
Ground Truth



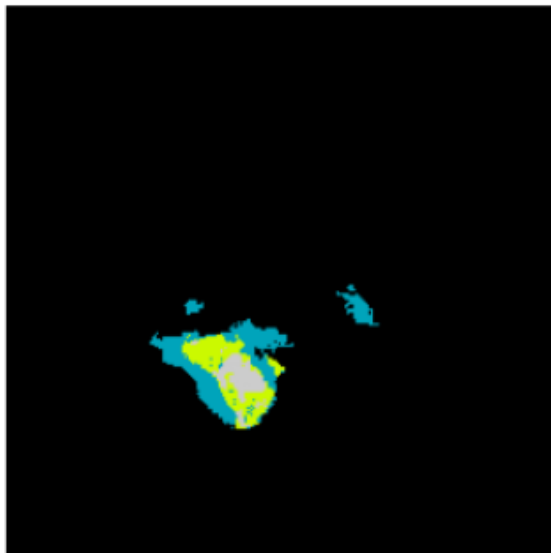
Prediction



Image



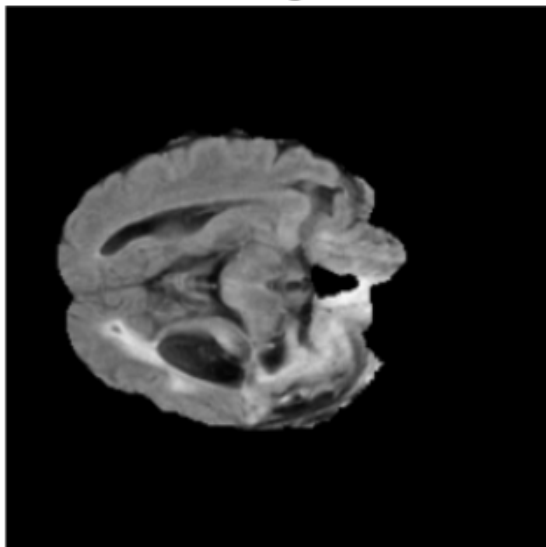
Ground Truth



Prediction



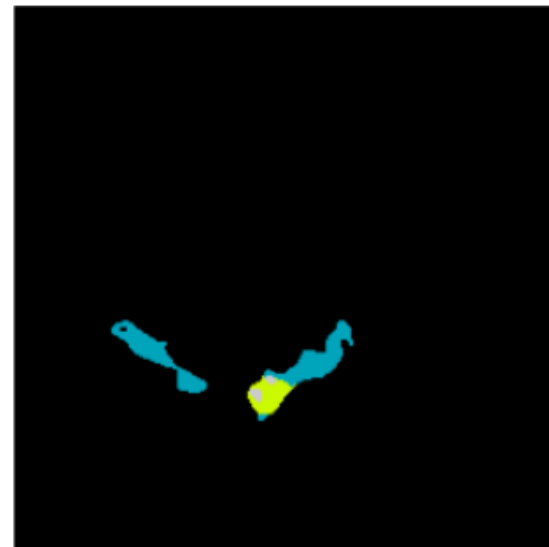
Image



Ground Truth



Prediction



FUTURE WORK

- Upgrade to 3D U-Net for volumetric context
- Use all modalities as multi-channel input
- Test other pretrained encoders (e.g., EfficientNet)

REFERENCES AND RESOURCES

- BRATS: <http://medicaldecathlon.com/>
- SMP: https://github.com/qubvel/segmentation_models.pytorch
- PyTorch: <https://pytorch.org>
- Nibabel: <https://nipy.org/nibabel>
- BRATS Paper: <https://www.nature.com/articles/s41467-022-30695-9>
- U-NET: <https://arxiv.org/abs/1505.04597>